Assumptions & Design Decisions

This dashboard has been developed by carefully aligning with the project requirements while also adapting to dataset limitations and improving analytical clarity.

1. Crop Type Slicer

The problem statement suggested including a Crop Type slicer. However, the dataset contains only rice-related metrics (Rice Area and Rice Production). Therefore, adding a crop type slicer would not provide any additional analytical value. Hence, it has been intentionally excluded.

2. Season Slicer

The dataset includes rainfall data only for monsoon months (June to September). Since explicit season categorization was not provided, a separate season slicer was not implemented. The analysis inherently focuses on monsoon-driven agricultural patterns.

3. Bar Chart Enhancement (Rainfall vs Yield)

The requirement mentioned bar charts comparing rainfall with crop yield. To address the difference in scale between rainfall and yield, a dual-axis visualization (Line and Clustered Column Chart) was used instead. This improves readability and enables meaningful comparison between the two measures.

4. Crop Yield Calculation

The dataset did not provide a direct yield column. Therefore, crop yield was derived using the formula:
Production ÷ Area.
This ensures accurate representation of agricultural productivity.

5. Data Cleaning Decisions

Certain values such as -1 and 0 in the Rice Area column were identified as invalid or missing data. These were removed to maintain data accuracy and ensure reliable analysis.

6. Focus on Key Metrics

To maintain a clean and user-friendly dashboard, only the most relevant KPIs were displayed. Yield-based metrics were prioritized over redundant area extremes to better represent productivity insights.

7. Use of Dynamic Insights

A dynamic insight panel was implemented using DAX to automatically update insights based on user selections (Region, State, Year). This enhances interactivity and decision-making.

Conclusion:

All design choices were made to ensure analytical accuracy, clarity, and better user experience while staying aligned with the intent of the problem

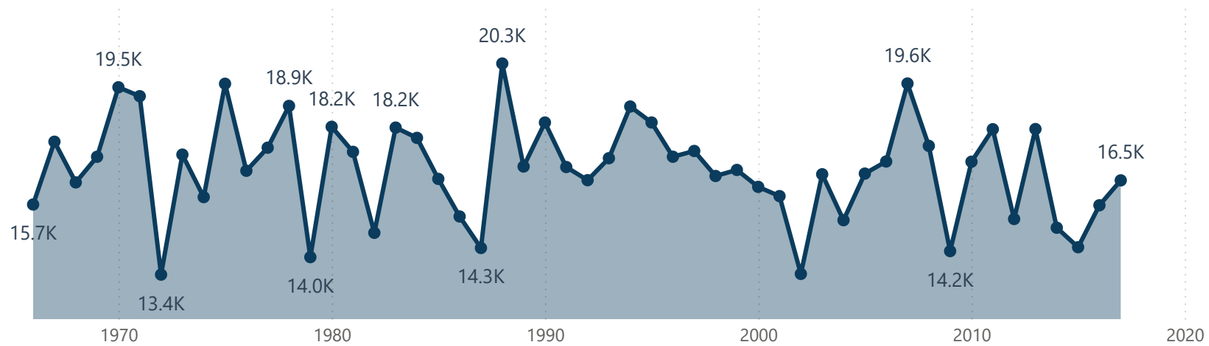
875.5K...	938.3 mm	142.7K ha	1.83	3.50	0.93
Total Rainfall	Avg Rainfall	Avg Rice Area	Avg Yield	Max Yield	Min Yield

Year: All

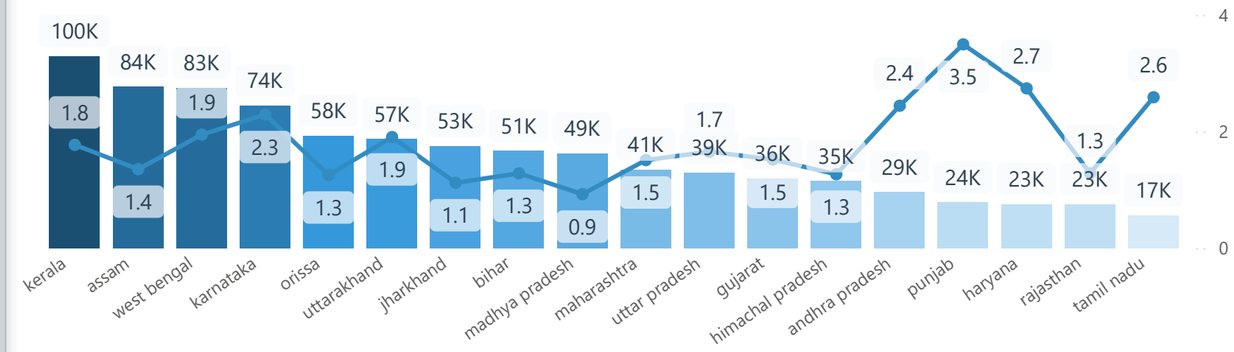
State Name: All

Region: All

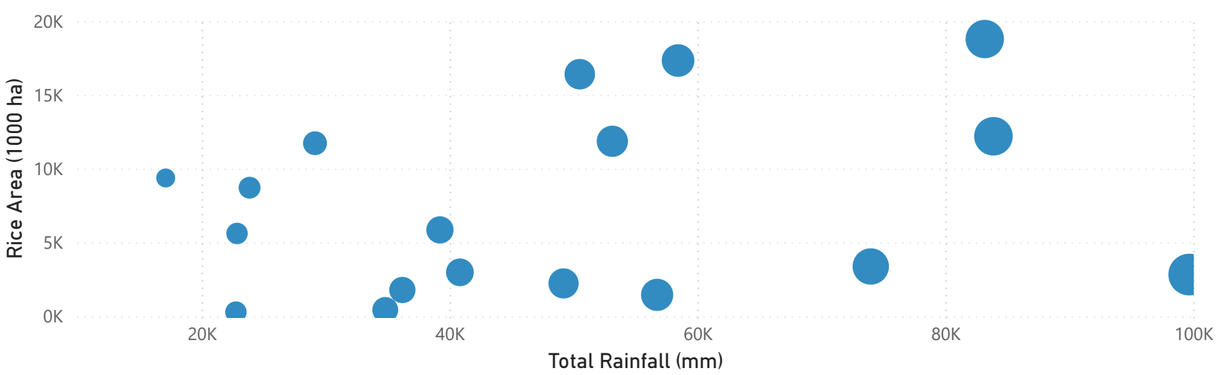
Year-wise Rainfall Trend (mm) (1966–2017)



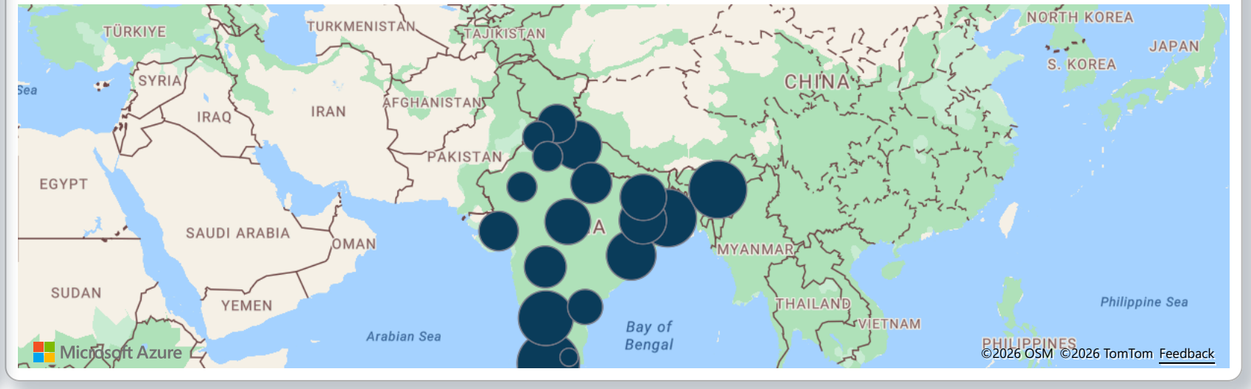
Rainfall (mm) vs Crop Yield by State



Rainfall (mm) vs Rice Area Correlation



State-wise Rainfall Distribution (mm)



📊 Insight: For All Regions, All States during All Years, total rainfall was 875.5K mm. Average rainfall was 938.3 mm, indicating high rainfall conditions.

Recommendations:

- ***Improve irrigation in low rainfall regions***
- ***Use water-efficient farming techniques***
- ***Optimize crop selection based on rainfall patterns***
- ***Develop drainage systems in high rainfall areas***